WNDRVR
SUPPORT NETWORK

VXWORKS 7 CREATING A CLI TOOL TUTORIAL, SR0650

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VxWorks 7 Creating a CLI Tool Tutorial, SR0650

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1. CLI TOOL CREATION OVERVIEW AND WORKFLOW

The VxWorks® 7 Creating a CLI Tool Tutorial helps you create a command-line interface tool to enable typed commands from a telnet shell or serial application.

In this example of external management, the end user uses a telnet client to connect to a telnet server and command interpreter embedded within the device. As with web-based management, the person managing the device can be on another part of the local network or the Internet.

This tutorial assumes that you have installed VxWorks and Workbench on your host system; that you are familiar with both VxWorks and Workbench. You should also have a basic familiarity with VxWorks 7 CLI, Web, MIBway. For more information, see:

- *VxWorks 7 CLI, Web, MIBway Programmer's Guide:* Overview
- *VxWorks 7 CLI, Web, MIBway Programmer's Guide:* Fundamentals

This tutorial does not make use of the management configuration editor (MCE). For additional tutorials, see VxWorks 7 Creating a Web Server Tutorial and VxWorks 7 Creating a CLI Tool and Web Server Tutorial.

Workflow

Perform the following workflow to creat the CLI tool.

> Note: This feature only works in kernel mode for VxWorks. It is not supported for RTP projects.

1. Create and configure a CLI VSB for the VxWorks Simulator (VxSim).
   See Creating a VSB for the CLI on page 2.
2. Create and configure a CLI VIP for VxSim.
   See Creating a VIP for the CLI on page 3.
3. Install the Wind River WRTAP network driver and configure network connectivity to enable IP communications between VxSim and your host.
   See Installing the Wind River TAP Driver on page 4.
4. Create a target connection between the simulator and your host.
   See Configuring a Network Connection to the Simulator on page 5.
5. Launch the VxSim daemon, vxsimnetd.
   See Starting the VxSim Network Daemon on page 7.
6. Launch the simulator instance and connect to the instance.
   See Launching the Simulator Instance and Connecting to the Simulator on page 8.
7. Access the CLI shell from a telnet session.
   See Accessing the CLI on page 9.

2. CREATING A VSB FOR THE CLI

You can create a VxWorks source build (VSB) project for a CLI tool running on a VxSim simulator on a Windows host.

Procedure

1. Base the VSB on the VxWorks simulator by selecting either **SIMNT** for the CPU or **vxsim_windows**_*x*_*x*_*x*_*x* for the BSP.
2. Enable the WEBCLI option.
3. Enable the WEBCLI_CLIDEMO option.
4. View the default configuration of the WEBCLI_COMMON parameters. You can retain the default configuration for the tutorial.
5. View the default configuration of the WEBCLI_CLI parameters. You can retain the default configuration for the tutorial.
6. Build the VSB.

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3. CREATING A VIP FOR THE CLI

You can create a VxWorks image project (VIP) to support a CLI tool running on a VxSim simulator on a Windows host.

Procedure

1. Base the VIP on the VSB created in Creating a VSB for the CLI on page 2.
2. Add the **INCLUDE_WEBCLI_CLIDEMO** component.
3. Build the VIP.

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4. INSTALLING THE WIND RIVER TAP DRIVER

To simulate network IP connectivity, VxSim includes a Wind River TAP (WRTAP) driver that allows VxWorks to communicate by handing off IP packets to the host operating system for processing.

About This Task

Once installed, the WRTAP driver is automatically initialized upon launching VxSim.

For more information on using the WRTAP driver, see the VxWorks 7 VxSim User's Guide.

To install the Wind River WRTAP driver, proceed as follows:

Procedure

1. Open a Windows command shell as an administrator.
2. In the command shell, use the following command, where *installDir* is the name of your VxWorks installation directory:

   ```
   C:\> installDir\vxWorks-7\host\x86-win32\bin\wrtapInstaller.exe install wrtap
   ```

   Observe the presence of a new network connection on your host. For example, **VMware Virtual Ethernet Adapter for VMnet1**.

5. CONFIGURING A NETWORK CONNECTION TO THE SIMULATOR

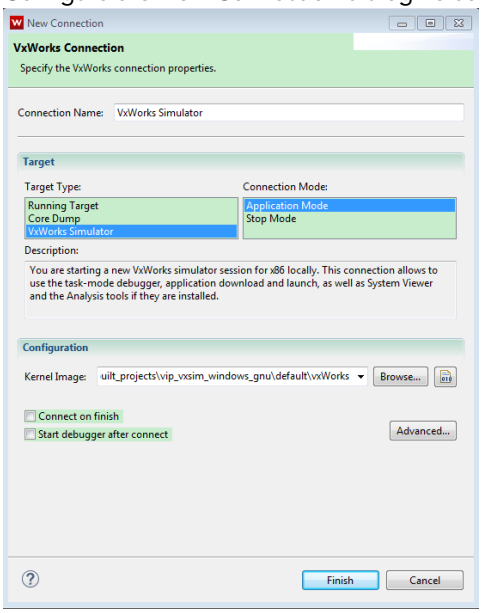
Before you can use the simulator, you must configure a network connection to it.

About This Task

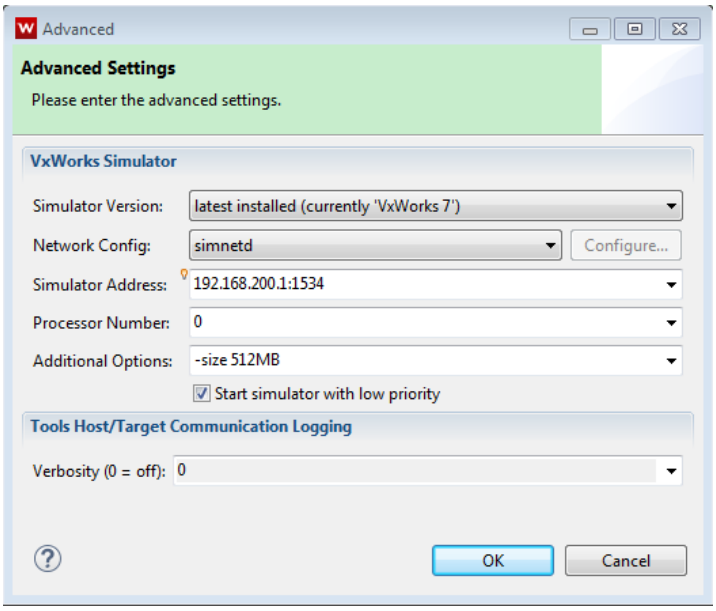
To configure a network connection between the simulator instance and your host, proceed as follows:

Procedure

1. From Workbench, select **New Connection**
2. Configure the New Connection dialog fields as follows:



3. Click Configure to display the Configure dialog.
4. Configure the fields as follows:



5. Click **Browse** and navigate to the CLI VxWorks image project (VIP) default folder and select the **vxWorks** executable.
6. Click **OK**.
7. From the New Connection dialog, click **Finish**.

6. STARTING THE VXSIM NETWORK DAEMON

If you use network services, you cannot start a VxWorks simulator instance until the VxWorks simulator network daemon is started and has completed its initialization.

About This Task

For more information on the VxSim network daemon, see the VxWorks 7 VxSim User's Guide.

To start the vxsimnetd daemon with the default configuration, proceed as follows:

Procedure

1. Open a Windows command shell as an administrator.
2. In the command shell, use the following command syntax, where *installDir* is the name of your VxWorks installation directory:

```
C:\> installDir\vxworks-7\host\x86-win32\bin\vxsimnetd
```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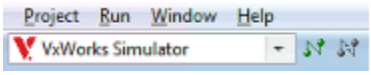
7. LAUNCHING THE SIMULATOR INSTANCE AND CONNECTING TO THE SIMULATOR

After configuring network connectivity, you can launch the simulator instance and connect to the simulator.

Procedure

1. If not already started, start the VxSim network daemon. See Starting the VxSim Network Daemon on page 7.
2. Select the **VxWorks Simulator** connection from the **Connection** drop-down list and click the **Connect** icon.



Terminal output similar to the following is displayed.

```
 _________          _________
 \77777777\        /77777777/
  \77777777\      /77777777/
   \77777777\    /77777777/
    \77777777\  /77777777/
     \7777777\  \7777777/
      \77777777\  \77777/            VxWorks 7
       \77777777\  \777/
        \77777777\  \7/     Core Kernel version: 1.0.2.0
         \77777777\  -      Build date: Jul  2 2014 16:13:16
          \77777777\
           \7777777/        Copyright Wind River Systems, Inc.
            \77777/  -                1984-2014
             \777/  /7\
              \7/  /777\
               -  -------

             Board: Windows 6.1
     OS Memory Size: 512MB
   ED&R Policy Mode: Deployed

Loading symbol table from host:C:/WindRiver_70_June_8_8B/workspace

->
```

3. From the VxWorks Simulator terminal, run the ifconfig command.

   Observe the presence of the 192.168.200.1 network interface. For example:

```
-> ifconfig
...
simnet0 Link type:Ethernet HWaddr 7a:7a:c0:a8:c8:01 Queue:none
inet 192.168.200.1 mask 255.255.255.0 broadcast 192.168.200.255
inet6 unicast fe80::787a:c0ff:fea8:c801%simnet0 prefixlen 64
...
```

8. ACCESSING THE CLI

After launching your kernel image and connecting to the simulator instance, you can access the CLI shell from a telnet session and execute commands.

About This Task

You can access the CLI shell as follows:

Procedure

1. Launch a terminal application such as Putty.
2. Configure a telnet session with the following parameters:
   - IP address: 192.168.200.1
   - Port: 2300
3. Login to the session with the following default credentials:
   - Login: admin
   - Password: admin

   Observe the presence of the CLI prompt. You now have an active telnet session with access to the CLI shell:

```
Login:admin
Password:
CLI_TEST#
```

9. GETTING HELP

Use the help command to list commands and help *commandName* to get help for a specific command.

Procedure

- To display the list of available commands, enter help at the command prompt.
- To display help for a specific command, use the following command syntax:

```
CLI_TEST# help commandName
```

For example:

```
CLI_TEST# help windmark
  windmark - test get, set and table windmark
  get - get sysName
  set - set sysName <string>
  table - table tcpConnTable
CLI_TEST# help windmark get
  get - get sysName
  sysName - absolute
CLI_TEST# help windmark set
  set - set sysName <string>
  sysName - string
```

10. USING COMMAND ALIASES

A command alias is a shortcut for a command.

About This Task

When you type the alias, the text is substituted automatically for the desired command.

Procedure

1. Use the alias command to create a new alias.

   Note: If your command contains spaces, begin and terminate it with double quotes.

   For example:

   ```
   CLI_TEST# alias name windmark
   ```

2. Test the alias.

   ```
   CLI_TEST# name
   sysName: AlphaOmega
   ```

11. ACCESSING THE VXWORKS KERNEL SHELL FROM THE CLI SHELL

You can access the VxWorks kernel shell from the CLI shell using the vxshell command.

Procedure

1. From the VxWorks Simulator terminal window, run the following initialization function:

```
-> webcliShellCmdInit()
value = 0 = 0x0
```

2. From the CLI shell window, run the following command and observe the presence of the VxWorks kernel shell prompt:

```
CLI_TEST# vxshell
->
```

3. To exit the VxWorks kernel shell and return to the CLI shell, enter the following command from the CLI shell window:

```
-> exit
Au revoir!
CLI_TEST#
```

12. ACCESSING THE CLI SHELL FROM THE VXWORKS KERNEL SHELL

You can access the CLI shell from the VxWorks kernel shell using the webcli command.

Procedure

1. From the VxWorks Simulator terminal window, run the following initialization function:

```
-> webcliShellCmdInit()
value = 0 = 0x0
```

2. Run the following command to access the VxWorks command-interpreter mode:

```
-> cmd
[vxWorks *]#
```

3. To access the CLI shell, enter the webcli command and login with the default credentials:

```
[vxWorks *]# webcli
Login: admin
Password:admin
CLI_TEST#
```

4. To logout from the CLI shell and return to the VxWorks kernel shell, enter the following command from the VxWorks Simulator terminal window:

```
CLI_TEST# logout
[vxWorks *]#
```

Note: Nested shell invocation is not supported.

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13. FILES USED IN THIS TUTORIAL

The files for the demonstration are located in *installDir* **/vxworks-7/pkgs_v2/app/webcli-***x.x.x.x* **/clidemo-***x.x.x.x* **/**.

In general, you should not edit these files beyond configuring them with Management Configuration Editor (MCE).

The following code files enable shell swapping between the VxWorks kernel shell and the CLI shell:

- *installDir* **/vxworks-7/pkgs_v2/app/webcli-***x.x.x.x* **/clidemo-x.x.x.x/src/cli2knl.c**

  This code file changes the command shell environment from the CLI shell to the VxWorks kernel shell. See Accessing the VxWorks Kernel Shell from the CLI Shell on page 12.
- *installDir* **/vxworks-7/pkgs_v2/app/webcli-***x.x.x.x* **/clidemo-x.x.x.x/src/knl2cli.c**

  This code file changes the command shell environment from the VxWorks kernel shell to the CLI shell. See Accessing the CLI Shell from the VxWorks Kernel Shell on page 13.